PHAM Van Tuan

GEOSPATIAL DATA SCIENTIST

PH.D. OF COMPUTER SCIENCE

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EDUCATION

2020 — 2023

Ph.D. of Computer Science

University of Lorraine, Metz, France

2014 — 2016

Master of Science in Information Technology

University of Lorraine & USTH, France

2008 — 2012

Bachelor of Science in Information Management Systems

National Economics University, Vietnam

PROFESSIONAL SUMMARY

- **Ph.D. in Computer Science** with over 8 years of cross-functional expertise bridging **Geospatial Data Science, Machine Learning and AI**.
- Proven track record of leading end-to-end R&D lifecycles—from complex predictive modeling to scalable full-stack deployments—delivering high-impact commercial solutions.

WORK EXPERIENCE

Geospatial Data Scientist — 10/2024 - PRESENT

Databourg-Komunidad Global, Luxembourg

- **Real-time Weather Forecasting:** Led development of multi-source forecast models, processing **60,000+ IoT/satellite sensors** natively. Increased accuracy by **5%** via Deep Learning modernization.
- **Infrastructure Modernization:** Re-architected pipelines via **Airflow, RabbitMQ, Docker**, automating data throughput and boosting system metrics by **20-40%**.
- **Full-Stack Web Deployment:** Delivered commercial GIS dashboard services (Django/FastAPI/MongoDB/MapboxGL) actively used by public agencies across **Philippines, Indonesia, Brazil**.
- **Climate Score Platform:** Engineered critical statistical frameworks analyzing **multi-terabyte netCDF metrics** for sustainable environmental strategies.

Postdoctoral / R&D engineer — 12/2023 - 06/2024

LCOMP - Univ. of Lorraine, Metz, France

- **Technical Leadership:** Spearheaded the core development phase of the **DCASOLVER** project as Tech-Lead, architecting and launching its integrated commercial web platform.
- **Industrial Optimization:** Engineered advanced mathematical algorithms solving complex business constraints, heavily focused on **GIS-based logistics and routing** applications.

PhD Candidate / R&D engineer — 10/2020 - 11/2023

University of Lorraine, Metz, France

- **Ph.D. Research:** Authored thesis "New machine learning techniques for finance and healthcare," proposing novel ML architectures to tackle complex domain challenges, including leveraging advanced NLP and fine-tuned Large Language Models (BERT/RoBERTa) to address critical health data scarcity.
- **Defense R&D Partnerships:** Executed 3 joint academic-industry research projects for a major defense contractor (NAVAL Group), applying Deep Learning, Computer Vision, and Robotics.

University Lecturer — 08/2012 - 10/2020

National Economics University, Hanoi, Vietnam

- Taught courses: Fundamental Computer Science, Applied Informatics, MIS.
- Conducted research in the domain of Artificial Intelligence.

GIS Data Engineer / Web Developer — 08/2019 - 12/2019

VinTech Group, Hanoi, Vietnam

- Engaged in the implementation of an AI-based Logistics system (web) with Django, Bootstrap, GoogleMap APIs for Vinmart's retail operations encompassing 2000 stores across Vietnam.
- Collected data, built the pipeline for processing data, designed the MySQL database, deployed to AWS cloud.

SKILLS HIGHLIGHTS

CORE AI & DS

Machine Learning | Geospatial AI | LLMs & NLP | Data Processing

GEOSPATIAL

GDAL/Rasterio | netCDF/HDF5 | MapboxGL | Spatial Analysis

DATABASE & INFRA

PostgreSQL/MySQL | MongoDB | Docker | Airflow/RabbitMQ

WEB & CLOUD

Django/FastAPI | RESTful APIs | AWS Cloud

LANGUAGES

Python (OOP) | C/C++ | JavaScript/TS

TOOLS

PyTorch/Tensorflow | Sklearn/Pandas | OpenCV | Git Flow

LANGUAGES

- **Vietnamese:** Native
- **English:** Professional
- **French:** Elementary

HOBBIES

- Running / Football
- Reading / Learning
- Taking care of family

ML Engineer / Product Owner

12/2017 - 07/2019

NAL Jsc., Hanoi, Vietnam

- Led a young and passionate team including 8 members driving R&D in the field of AI.
- Developed AI-based applications for various outsourcing projects: Operator Chatbot, Iris recognition & tracking, Face lock System, Dialog System, and Hate Speech Detection.

NLP Engineer

10/2016 - 10/2017

Chappiebot Inc., Hanoi, Vietnam

- Developed clever algorithms and pragmatic solutions for an AI-Based Car Search System.
- Implemented NLP tasks: NER, Topics classifiers, Sentiment Analysis, and Dialogue Systems in Vietnamese.

ML Research Intern

04/2016 - 10/2016

LITA - Univ. of Lorraine, Metz, France

- Thesis: "Machine learning techniques for Autonomous Surface Vessels".
- Conducted research on long-term autonomy and data acquisition for environmental monitoring using Reinforcement learning, robotics control techniques on Unmanned Surface Vehicles (USV).

KEY PROJECTS

- **Global Climate Risk System (2025):** Platform processing multi-terabyte climate datasets. *(GDAL/Rasterio, QGIS, netCDF/TIF, Machine learning/Statistics, Python, React)*.
- **Real-time Rainfall Forecasting (2024):** ML pipelines for multi-source precipitation predicting. *(TensorFlow, GDAL/Rasterio, FastAPI, Airflow, Python, Django)*.
- **DCASOLVER Platform (2024):** Tech-Lead for logistics route optimization platform. *(Python, Django, FastAPI, MySQL, Linux)*.
- **Robot navigation for NAVAL Group (2020):** Ground robot path optimization. *(Computer vision, YOLO, ROS, Gazebo, SLAM)*.
- **Iris recognition & tracking (2018):** Tracking user behavior on mobile devices. *(Computer Vision, YOLO, RestAPI)*.
- **Face lock System (2018):** Automated check-in and timekeeping. *(Deep Learning, Few-shot Learning)*.
- **Automated hate speech detection (2017):** Sentiment classification for Japanese client. *(FastText, Python, Flask)*.
- **Customer service chatbot (2017):** Automated CSR for Thaiminh Group. *(Deep Learning, Text-To-Speech, Docker)*.
- **Search-based NLP in Otonhanh.vn (2016):** Automobile search platform. *(ML, NLP, Scikit-learn)*.

PUBLICATIONS

- Pham, V.T., Luu, H.P.H., Le Thi, H.A. (2022). **A Block Coordinate DCA Approach for Large-Scale Kernel SVM**. ICCCI 2022.
- Pham, V.T., Le Thi, H.A., Luu, H.P.H., Damel, P. (2023). **DCA-Based Weighted Bagging: A New Ensemble Learning Approach**. ACIIDS 2023.
- Pham, V.T., Le Thi, H.A. and Pascal, D. (2023). **Cost-sensitive weighted bagging DCA based method for imbalanced financial data**. Submitted to NUMTA.
- Pham, V.T. (2023). **Techniques d'apprentissage automatique basées sur DCA**. Ph.D. Thesis Publication.